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APPLICATION - THIS STANDARD IS APPLICABLE TO NAVY OR MARINE CORPS REBUILT TIRE CONTRACTS THAT SPECIFY FLIGHT TESTING TO DEMONSTRATE SUITABILITY OF PRODUCTION REBUILDING FOR CARRIER ENVIRONMENTS.

PURPOSE - THIS STANDARD ESTABLISHES FLIGHT TESTING REQUIREMENTS FOR QUALIFIED (MIL-R-7726) HIGH SPEED/HIGH PERFORMANCE REBUILT AIRCRAFT TIRES.

REQUIREMENTS -

Ⓐ ~~1. SAMPLE SELECTION. TWELVE (12) TIRES SHALL BE RANDOMLY SELECTED FROM THE FIRST ONE HUNDRED (100) PRODUCED BY THE CONTRACTOR. THE TIRES SHALL BE FORWARDED TO THE TESTING FACILITY FOR USE AS REQUIRED TO DETERMINE CONFORMANCE TO THIS STANDARD.~~

2. THE REBUILT TIRES SHALL BE TESTED IN EACH OF FOUR (4) FLIGHT TEST ENVIRONMENTS (PHASES). PHASES 1, 2, AND 3 ARE MANDATORY REQUIREMENTS. PHASES 4 SHALL BE AT THE OPTION OF THE TESTING ACTIVITY.

NOTES: FOR ENTIRE TEST SPECTRUM:

1. THE HIGH-SPEED/HIGH PERFORMANCE REBUILT TEST TIRES SHALL BE SERVICED, INSPECTED AND MAINTAINED IN ACCORDANCE WITH TECHNICAL MANUAL NAVAIR 04-10-506, APPLICABLE MAINTENANCE MANUALS AND LATEST DIRECTIVES.
2. TIRE PRESSURES SHALL BE CHECKED PRIOR TO THE FIRST FLIGHT OF THE DAY WITH A SERVICEABLE, CALIBRATED GAGE.
3. TREAD DEPTH (WEAR) MEASUREMENTS SHALL BE TAKEN AT A MINIMUM OF EVERY TWO FLIGHTS.
4. TIRE CHANGE REQUIREMENTS ON ALL TIRES INSTALLED ON THE TEST AIRPLANE WHILE PERFORMING HIGH-SPEED/HIGH PERFORMANCE REBUILT TIRE TESTS, SHALL BE SUBJECT TO THE TEST ENGINEER'S DISCRETION.
5. THE TEST REQUIREMENTS OUTLINED ARE MINIMUM REQUIREMENTS. ADDITIONAL TESTS MAY BE CONDUCTED AT THE TESTING FACILITY'S DISCRETION IF PROBLEM AREAS ARE DISCOVERED.
6. ALL TESTS SHALL BE CONDUCTED ON DRY SURFACES.

(A) PHASE ONE - MAXIMUM GROSS WEIGHT CATAPULT LAUNCHES WITH MINIMUM SINK RATE FIELD LANDING

CATAPULT LAUNCH TEST CONDITIONS

LAUNCH NO.(1)	AIRPLANE GROSS WEIGHT (LBS)(2)	OFF-CENTER SPOTTING NOSE/MAIN(3)&(4)	HORIZONTAL TOW FORCE (LBS)	AIRPLANE LONGITUDINAL ACCELERATION (g)
1	AS REQ.	0/0	AS REQ.	AS REQ.
2	AS REQ.	0/0	AS REQ.	AS REQ.
3	MAX.	0/0	MAX.	AS REQ.
4	MAX.	6L/0	MAX.	AS REQ.
5	MAX.	6R/0	MAX.	AS REQ.
6	MAX.	6L/6L	MAX.	AS REQ.
7	MAX.	6R/6R	MAX.	AS REQ.
8	MAX.	0/6L	MAX.	AS REQ.
9	MAX.	0/6R	MAX.	AS REQ.
10	AS REQ.	0/0	MAX.	MAX.

PHASE 1 NOTES:

(1) LAUNCH NUMBERS (1) & (2) ARE OPTIONAL BUILDUP LAUNCHES AND SHALL BE CONDUCTED AT A REDUCED GROSS WEIGHT DETERMINED BY THE TESTING ACTIVITY.

(2) MAXIMUM GROSS WEIGHT - 95% TO 100% MAXIMUM.

(3) NOSE/MAIN GEAR SPOTTING - INCHES OFF CENTERLINE.

(4) NOSE GEAR, OFF CENTER SPOTTING IS NOT PERMITTED ON NOSE TOW AIRPLANES.

(5) ALL LAUNCHES SHALL BE FOLLOWED BY MINIMUM SINK RATE FIELD LANDING AT MINIMUM CROSSWIND CONDITIONS AND WITH MINIMUM BRAKING DURING LANDING ROLLOUT.

APPROVED 11 JAN 1976 REVISED Ⓐ 10 JUNE 1983

REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

Ⓐ DENOTES CHANGES

This military standard is approved by Naval Air Systems Command, [illegible]

F.A. NAVY - AS Other Cust	TITLE SERVICE SUITABILITY (FLIGHT) TESTING OF REBUILT NAVY AIRCRAFT TIRES	MILITARY STANDARD MS14147(AS)
PROCUREMENT SPECIFICATION	SUPERSEDES	SHEET 1 OF 3

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(B) PHASE TWO - HEAVY GROSS WEIGHT CATAPULT LAUNCHES WITH MAXIMUM GROSS WEIGHT ARRESTED LANDINGS. THE CATAPULT SCHEDULE SHALL BE CONDUCTED AS PRESENTED FOR PHASE ONE EXCEPT AS LISTED BELOW:

1. AIRPLANE GROSS WEIGHT FOR LAUNCH NUMBERS 1, 2, AND 3 SHALL VARY BY EVEN INCREMENTS BETWEEN MAXIMUM ARRESTED LANDING GROSS WEIGHT PLUS 5% TO MAXIMUM PRACTICAL GROSS WEIGHT AVAILABLE.

2. CATAPULT LAUNCH WEIGHTS FOR LAUNCH NUMBERS 4 THROUGH 10 SHALL BE SELECTED BY THE TESTING FACILITY AND SHALL BE GREATER THAN THE MAXIMUM ARREST LANDING WEIGHT PLUS 5%. WEIGHT SELECTIONS SHALL BE MADE BASED UPON GATHERED DATA, BUT WILL ALLOW FOR A MINIMUM OF 4 MAXIMUM LONGITUDINAL ACCELERATION CATAPULT LAUNCHES DURING OFF-CENTER SPOTTING CONDITIONS. UNLESS RESTRICTED BY AIRFRAME LIMITATIONS, THE ARRESTED LANDINGS FOR PHASE TWO SHALL FOLLOW THE SCHEDULE BELOW. THESE LANDINGS SUPERSEDE THOSE OF NOTE (5) OF THE CATAPULT LAUNCH TEST CONDITIONS.

ARRESTED LANDING TEST CONDITIONS

LANDING NO.	TYPE LANDING (2)	GROSS WEIGHT (LBS)(1)	ENGAGING SPEED (FT)	SINK SPEED (fps)	AIRCRAFT LONG ACCEL (-g)
1	NORMAL	AS REQ.	AS REQ.	AS REQ.	AS REQ.
2	NORMAL	MAX.	AS REQ.	AS REQ.	AS REQ.
3	NORMAL	MAX.	AS REQ.	80%-100%	AS REQ.
4	ROLL/YAW SAME	MAX.	AS REQ.	AS REQ.	AS REQ.
5	ROLL/YAW OPPOSITE	MAX.	AS REQ.	AS REQ.	AS REQ.
6	OFF CENTER LEFT	MAX.	AS REQ.	AS REQ.	AS REQ.
7	OFF CENTER RIGHT	MAX.	AS REQ.	AS REQ.	AS REQ.
8	FREE-FLIGHT	MAX.	AS REQ.	AS REQ.	AS REQ.
9	NORMAL	AS REQ.	AS REQ.	AS REQ.	MAX.
10	NORMAL	AS REQ.	MAX.	AS REQ.	AS REQ.

PHASE 2 NOTES: (1) MAXIMUM GROSS WEIGHT - 95% TO 100% MAXIMUM.

(2) PERFORMANCE OF THE TEST CONDITIONS ARE SUBJECT TO THE FLIGHT RESTRICTIONS AND LIMITATIONS OF THE TEST AIRCRAFT.

(C) PHASE THREE - MAXIMUM GROSS WEIGHT FIELD TAKEOFFS AND LANDINGS. MINIMUM TEST REQUIREMENTS, WITH APPLICABLE LIMIT TEST CONDITIONS, ARE PRESENTED BELOW:

1. CROSSWIND TESTING. A MINIMUM OF TWO MAXIMUM GROSS WEIGHT CROSSWIND TAKEOFFS AND LANDINGS (BOTH LEFT AND RIGHT) SHALL BE CONDUCTED AT THE MAXIMUM CROSSWIND CONDITION AVAILABLE (NOT TO EXCEED 85% OF THE AIRPLANE LIMITATION).

2. MAXIMUM ROTATION SPEED. A MINIMUM OF TWO 95% TO 100% MAXIMUM ROTATION SPEED TAKEOFFS SHALL BE CONDUCTED AT 95% TO 100% MAXIMUM TEST AIRPLANE TAKEOFF GROSS WEIGHT.

3. MAXIMUM LANDING SPEEDS. A MINIMUM OF TWO 95% TO 100% MAXIMUM AIR SPEED, MINIMUM SINK RATE, MINIMUM CROSSWIND AND MINIMUM BRAKING LANDING SHALL BE CONDUCTED AT TEST AIRPLANE MAXIMUM GROSS WEIGHT FIELD LANDING CONDITIONS.

4. FIELD CARRIER LANDING PRACTICE. A MINIMUM OF ONE FIELD CARRIER LANDING PRACTICE (FCLP) CYCLE SHALL BE CONDUCTED. THIS CYCLE SHALL CONSIST OF NOT LESS THAN SIX FCLP TOUCH AND GO LANDINGS, STARTING AT MAXIMUM FCLP LANDING WEIGHT.

PHASE 3 NOTE: MAXIMUM GROSS WEIGHT - 95% TO 100% MAXIMUM.

(D) PHASE FOUR - (IF NECESSARY) REBUILT TIRE WEAR TESTING.

1. REBUILT TIRE WEAR TESTING. ADDITIONAL MAXIMUM AIRPLANE GROSS WEIGHT FIELD TESTING SHALL BE CONDUCTED IF EXCESSIVE WEAR IS ENCOUNTERED DURING ANY PHASE OF THE HIGH-SPEED/HIGH PERFORMANCE REBUILT TIRE TESTING. THIS TEST SHALL BE A SIDE-BY-SIDE EVALUATION UTILIZING A CONTRACT REBUILT TIRE AND A STANDARD NEW PRODUCTION TIRE. THE TEST REQUIREMENTS SHALL BE A MINIMUM OF TEN MAXIMUM GROSS WEIGHT FIELD TAKE-OFFS AND INTERMEDIATE GROSS WEIGHT FIELD LANDINGS TO BE CONDUCTED ON DRY CONCRETE SURFACES DURING MINIMUM CROSSWIND CONDITIONS.

PHASE 4 NOTES: (1) MAXIMUM GROSS WEIGHT - 95% TO 100% MAXIMUM.

(2) INTERMEDIATE GROSS WEIGHT - ANY GROSS WEIGHT BETWEEN THE MAXIMUM GROSS FIELD LANDING WEIGHT AND MAXIMUM FIELD CARRIER LANDING PRACTICE LANDING WEIGHT.

P.A. NAVY - AS Other Cust	TITLE SERVICE SUITABILITY (FLIGHT) TESTING OF REBUILT NAVY AIRCRAFT TIRES	MILITARY STANDARD MS14147(AS)
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3. REPORTING OF DATA - THE TESTING FACILITY SHALL PREPARE REPORTS AS FOLLOWS:

A. INITIAL LETTER REPORT - PREPARED TO INDICATE RECEIPT OF TIRES AND INITIATION OF TESTING

B. UNSATISFACTORY PERFORMANCE REPORT - LETTER OR MESSAGE AS APPROPRIATE

C. FINAL REPORT - FORMAL TEST REPORT

APPROVED 13 JAN 1976 REVISED (A) FOR CHANGES SEE SHEET 1

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P.A. NAVY - AS
Other Cust

TITLE
SERVICE SUITABILITY (FLIGHT) TESTING OF
REBUILT NAVY AIRCRAFT TIRES

MILITARY STANDARD
MS14147(AS)

PROCUREMENT SPECIFICATION

SUPERSEDES

SHEET 3 OF 3

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